DGT: Deformable Graph Transformer for Hyperspectral Image Classification

Yingjie Lu, Xiaofei Wang, *Graduate Student Member, IEEE*, Shaohui Mei, *Senior Member, IEEE*, Fulin Xu, and Mingyang Ma, *Graduate Student Member, IEEE*

***Abstract*— Transformers can model global context to enhance the performance of hyperspectral classification. However, the explored global information is generally confined to the spatial neighborhood of target pixels. In order to fully leverage global correlation across broader areas, a deformable graph transformer (DGT) is proposed for hyperspectral classification, in which the global information within an entire image is explored to improve the classification performance. Specifically, DGT layers are designed to adaptively sample virtual nodes at varying distances from an initial graph constructed from an image, by which the global spatial information can be explored using a deformable graph self-attention (DGSA) mechanism. Moreover, a learnable absolute position encoding (LAPE) module is constructed to enhance the spatial context awareness of DGT by integrating positional information into the graph nodes. In addition, graph structure encoding and graph topology encoding are further designed as inductive biases for the graph, by which both local structural information and global topological information of the HSI are captured to enhance the feature extraction capability of the DGT layer. Ultimately, through the stacking of multiple DGT layers, a composite feature fusion learning (CFFL) module is employed to fully utilize the simple low-level and complex abstract high-level features extracted from different layers. Extensive experiments on four datasets demonstrate the superiority and robustness of the proposed DGT over several state-of-the-art (SOTA) methods in terms of various evaluation criteria.**

***Index Terms*— Deformable graph self-attention (DGSA), graph structure and topology encoding, graph transformer, hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL image (HSI) is composed of hundreds of near-continuous spectral bands, which provides rich high-resolution spectral details and spatial distributions of diverse ground objects. Due to the differing reflectance properties of different materials across electromagnetic wavelengths in nature, it can reflect the physical and chemical composition of various substances. HSI classification task, on the other hand, takes advantage of this characteristic to realize the classification of each pixel in hyperspectral data, which currently plays a great role in environmental monitoring [1], agricultural production [2], mineral detection [3], and other fields.

In the early stages of development, some machine learning-based methods were applied to HSI classification [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. However, these methods often necessitate the manual design of features, rendering it challenging to mine deep nonlinear spectral–spatial information. With advancements in technology, deep learning methods have been applied to HSI classification tasks [15], [16], [17], [18], [19], and many CNN-based methods emerged [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30]. Inspired by human perception, attention mechanisms have been proposed to enhance the model's focus on different parts of the data [31]. A special variant of the attention mechanism, the self-attention (SA) mechanism, can consider information from all positions in the input sequence simultaneously. This powerful global modeling capability has been integrated into the transformer, which can capture global dependencies better than most deep learning-based models [32].

Since the introduction of the vision transformer (ViT) [33], which first applied pure transformers in the field of computer vision, numerous transformer-based methods have been applied to HSI classification tasks. He et al. [34] proposed a classification framework called HSI-BERT, which can dynamically and flexibly select different regions to input into the transformer encoder to capture global contextual information. Subsequently, Zhong et al. [35] introduced a spectral–spatial transformer network (SSTN), which broke the geometric constraints of the convolution operation. Hong et al. [36] approached the original HSI spectral bands from a sequential perspective, dividing them into groupwise spectral embeddings and learning local sequence information between adjacent bands through the proposed spectral former (SF). Sun et al. [37] utilized CNN to extract low-level spatial and spectral features from HSIs, which were then fed into a transformer encoder to extract high-level semantic information for pixel-wise classification. Furthermore, several other transformer architectures have been proposed for HSI classification, such as spatial–spectral transformer (SST) [38], group-aware hierarchical transformer (GAHT) [39], multiscale and cross-level attention learning (MCAL) network [40], graph generative structure-aware transformer (GraphGST)

Received 19 June 2024; revised 4 August 2024 and 21 September 2024; accepted 26 September 2024. Date of publication 8 October 2024; date of current version 18 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62171381 and Grant 62201445 and in part by the Project funded by China Postdoctoral Science Foundation under Grant 2022M722550. *(Corresponding author: Shaohui Mei.)*

The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: luyingjie@mail.nwpu.edu.cn; wangxiaofei2022@mail.nwpu.edu.cn; meish@nwpu.edu.cn; xfl@mail.nwpu.edu.cn; mamingyang@mail.nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3476327

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[41], cross-attention fusion transformer (CAF-Former) [42], and HSI transformer (HiT) [43].

In recent years, graph neural networks (GNNs) represented by graph convolutional networks (GCNs) have been widely applied in the field of HSI classification with their special ability to handle irregular graph data [44], [45]. Mou et al. [46] constructed a nonlocal graph based on the spectral similarity between pixels and proposed a network consisting of two stacked GCN layers to facilitate information propagation between each pixel and all other pixels. However, the graph constructed by using all pixels as graph nodes leads to an unbearable computational burden. To address this issue, Hong et al. [47] proposed miniGCN, which constructs multiple small-scale graphs as inputs to the GCN network. Wan et al. [48], on the other hand, significantly reduced the number of nodes in the graph by treating homogeneous regions composed of superpixels as nodes of the graph. Subsequently, many other GNN-based methods, such as graph attention networks (GATs), have been utilized to process graph structures formed by hyperspectral data [49], [50], [51]. Zhao et al. [52] embedded graph transformer into GCN to generate graphs that are then processed by GAT aggregating information from neighboring nodes with different importance to update the features of central nodes. Shi et al. [53] used the SA mechanism to assign different weights to neighboring nodes, which is similar to the mechanism of GAT.

Despite the satisfactory performance achieved by the aforementioned deep learning methods in the field of HSI classification, they still encounter several significant challenges, including the following.

1) Most methods based on CNN and transformer divide the original HSI into patches that are input to the model. These patches are 3-D cubes composed of pixels centered on training or test samples and their neighbors, each of which is independent of the other, which makes the model ignore the relationship between instances.
2) Constrained by the patch size, the SA mechanism of the transformer is only limited to the inside of the patches, which makes the model unable to capture long-distance dependencies despite the powerful global modeling capability of the SA mechanism.
3) Applying SA to all nodes in the graph can lead to oversmoothing issues, while focusing only on neighboring nodes ignores the global dependencies of the nodes composed of pixels. In addition, although Zhao et al. [52] applies graph transformer to the field of HSI classification, it still relies to some extent on GCN and GAT.

To address the aforementioned issues and explore the dependencies between distant pixels in HSIs, we propose a deformable graph transformer (DGT) classification framework. This framework integrates the advantages of GNNs in handling irregular data with the global modeling capabilities of transformers. Specifically, the original HSI is first fed into a learnable absolute position encoding (LAPE) module, where the regular data are encoded into non-Euclidean graph data. The encoded graph is then embedded into a positional space, ensuring that each node within the graph possesses its own positional information. This positional information serves a dual purpose: on the one hand, it is converted into positional embeddings that are incorporated into the graph node features; on the other hand, it is utilized by the DGT layer to adaptively select virtual nodes at various locations and distances as inputs for the encoder. The deformable graph SA (DGSA) mechanism, as the core component of DGT, integrates the initial graph structure encoding and the initial graph topology encoding, which serve as inductive biases to enhance the graph representation capabilities of DGT. Following the stacking of multiple DGT layers, the semantically nuanced information extracted from different levels, along with a composite of local and global information, is fed into a composite feature fusion learning (CFFL) module for further processing. The refined features learned ultimately are utilized to predict the categories of hyperspectral pixels.

The primary contributions of this article are specified as follows.

1) A graph transformer algorithm named DGT is proposed for HSI classification by selecting a fixed number of virtual neighbor nodes adaptively through DGSA, by which the dependencies between the global relation within an image are captured for the classification of pixels.
2) Graph structure encoding and graph topology encoding are specifically tailored for the DGSA, which enables the model to simultaneously consider both local structural information and global topological information of an HSI.
3) An LAPE module is proposed to facilitate the acquisition of positional embeddings for each node, which improves the ability of the DGT to recognize different regions within HSI by capturing the positional correlation between them.
4) A CFFL module is proposed, wherein complex abstract information and simple detail features from different layers are adaptively integrated and selected to enhance feature discriminability.

The rest part of this article is organized as follows. The structure of our proposed DGT is introduced in section II. Section III details a comprehensive experimental assessment of the proposed DGT model. Section IV provides the discussions of DGT. Conclusion is exhibited in Section V.

II. Proposed Method

A. Architecture of the Proposed DGT

The overall architecture of the proposed DGT is depicted in Fig. 1, comprising three main components: an LAPE module, DGT layers, and a CFFL module. Initially, the input HSI is encoded into an irregular graph structure by the LAPE module and embedded into positional space. This process provides positional information to the graph's nodes and converts this information into positional embeddings that are incorporated into the input sequence as positional encoding. Subsequently, these graph node sequences are processed by the DGT layers, composed of deformable graph attention mechanisms to extract global information from the HSI. This approach transcends the local constraints of patches and

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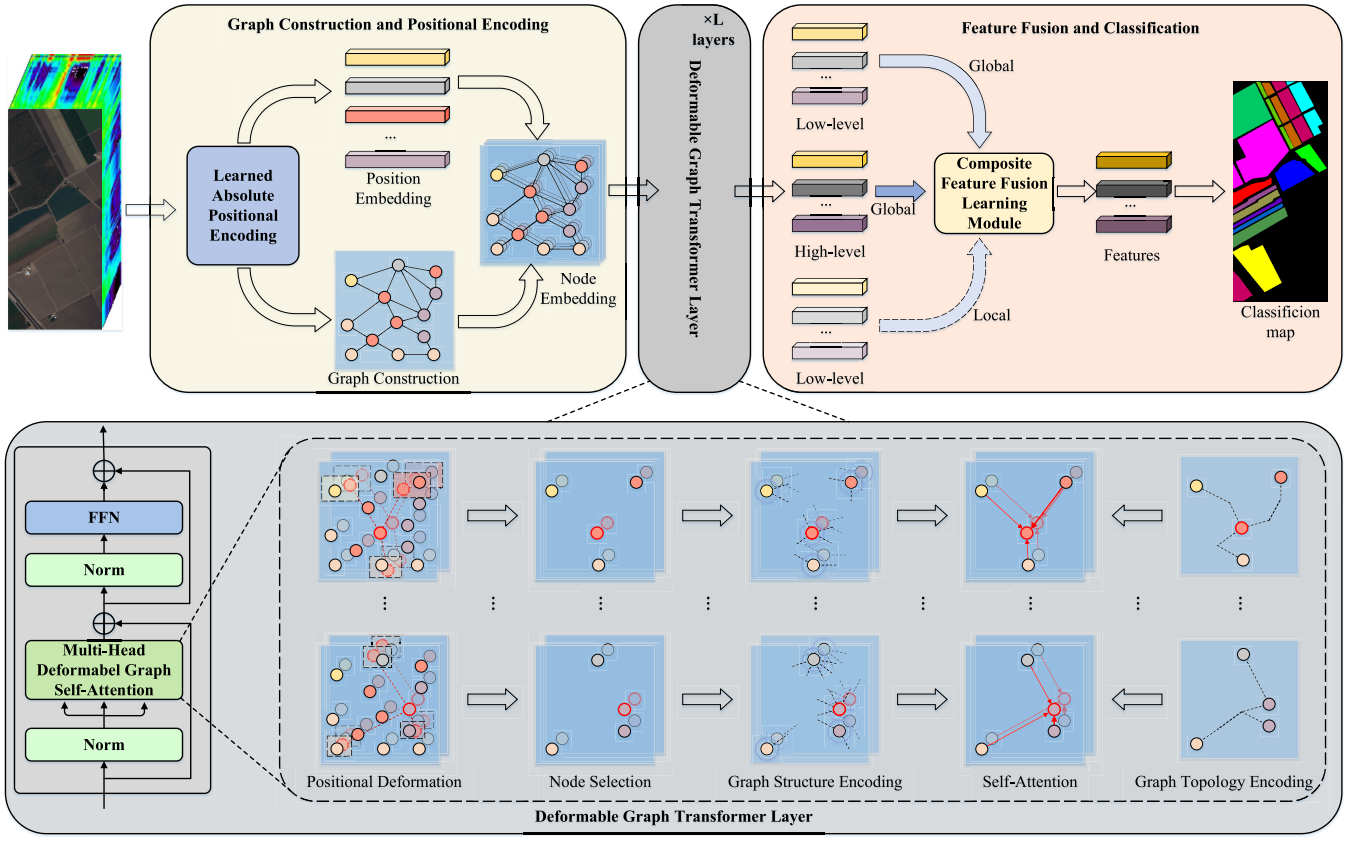


Fig. 1. Framework of our proposed DGT. It contains three main components: learnable absolute positional encoding, DGT layer, and CFFL.

neighboring node areas, enabling information exchange not only between distant nodes but also among adjacent ones. Finally, the complex high-level semantic information, along with simple low-level and both global and local information produced by different layers, is fed into a CFFL module for adaptive integration, which generates more comprehensive and expressive features for the precise classification of HSIs. These pivotal modules of the proposed method will be detailed further in Sections II-B–II-E.

B. Learnable Absolute Position Encoding

The primary objectives of the LAPE module are twofold. First, it encodes the raw HSI into a graph structure and subsequently maps this graph into a positional space, providing each node with unique positional information, which serves as a spatial foundation for the deformable layer. Second, the module transforms these encoded absolute positions into linear space, generating node embeddings for each graph node, which serve as the positional encodings for the DGT input node sequence. Fig. 2 details the intricate processes of this module.

For the input $\mathbf{X} \in \mathbb{R}^{h\times w\times c}$, it can be split into n pixel vectors $\boldsymbol{x}_i \in \mathbb{R}^{1\times c}$ along the spatial dimension, where h, w, and c represent the height, width, and number of channels, respectively, and $n = h \times w$. Initial processing involves pixel-level position encoding, where each pixel is encoded along two dimensions and the results are stored in $\mathbf{A}_p$,

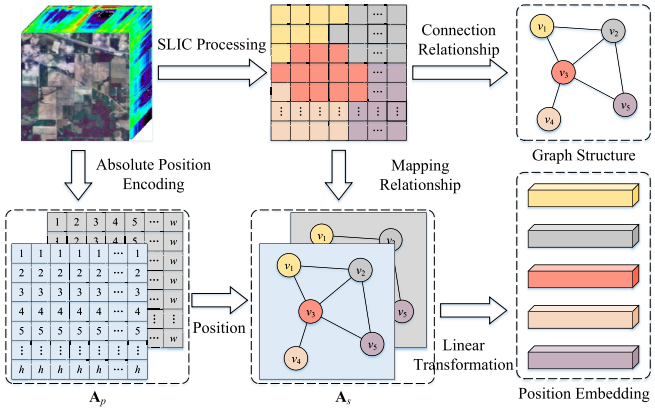


Fig. 2. Flowchart of the LAPE.

with $\mathbf{A}_p \in \mathbb{R}^{h\times w\times p}$, where p represents the number of channels after encoding. Fig. 2 provides specific examples of these dimensions. Following pixel-level encoding, the HSI is segmented into superpixels using the simple linear iterative clustering (SLIC) algorithm, with the segmentation scale set to s. This results in the creation of superpixels S and a pixel-to-superpixel mapping relationship matrix $\mathbf{M}$, where $\mathbf{M} \in \mathbb{R}^{hw\times b}$. The specific formula is given as follows:

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } \boldsymbol{x}_i \in S_j \\ 0, & \text{if } \boldsymbol{x}_i \notin S_j \end{cases} \tag{1}$$

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where $b = \lceil hw/s \rceil$, S_j is the jth superpixel, and $j \in \{1, 2, \ldots, b\}$. Based on $\mathbf{A}_p$ and $\mathbf{M}$, the absolute position encoding $\mathbf{A}_s$ for the superpixels is derived by the following expression:

$$\mathbf{A}_s = \overline{\mathbf{M}}^{\mathrm{T}} \mathrm{Reshape}\left(\mathbf{A}_p\right) \tag{2}$$

$$\overline{\mathbf{M}}_{ij} = \frac{\mathbf{M}_{ij}}{\sum_{k=1}^{n} \mathbf{M}_{kj}} \tag{3}$$

where Reshape$(\cdot)$ denotes the process of transforming a 3-D tensor into a 2-D matrix. Although the position encoding obtained provides location information for graph nodes, it cannot be directly integrated into the input node sequence. Consequently, we employ a learnable weight matrix $\mathbf{W}_p$ to linearly transform the absolute position encoding. The transformation is expressed as follows:

$$\mathbf{P} = \mathbf{A}_s \mathbf{W}_p \tag{4}$$

where $\mathbf{P}$ represents the generated positional embeddings that have the same dimensional size as each input node's embedding vector. These embeddings are then incorporated into the input node sequence to provide positional information. The expression is given as follows:

$$\mathbf{H}^0 = \mathbf{H} + \mathbf{P} = \overline{\mathbf{M}}^{\mathrm{T}} \mathrm{Reshape}\left(\mathbf{X}^o\right) + \mathbf{P} \tag{5}$$

where $\mathbf{X}^o \in \mathbb{R}^{h \times w \times d_0}$ is the output of the initial HSI after passing through two spectral convolutional layers, $\mathbf{H}$ is the output of the node embedding generator, and the structural information of the graph is stored in the adjacency matrix $\mathbf{A}$.

The introduced LAPE module strategically embeds the graph nodes within a defined positional space, endowing each node with unique positional data. This enhancement significantly bolsters the model's ability to discriminate among graph nodes.

C. Deformable Graph SA

The inputs to the DGSA consist of the graph nodes $\mathbf{H}^{(0)}$ and their position spaces $\mathbf{A}_s$. For each node, also called the central node, N position offsets are initially generated within a deformable region, where N denotes the number of neighboring nodes selected. This will induce a displacement transformation of the central node from its original position, thereby creating new nodes in the positional space, referred to as virtual neighbor nodes v'_j, where $j \in \{1, 2, \ldots, N\}$, and the hyperparameter λ is utilized to regulate the scale of the sampling range. These virtual nodes possess only absolute position information, lacking any feature information; thus, they cannot be directly processed by SA mechanisms. To ensure that virtual neighbor nodes, after positional deformation, represent other nodes within the graph at varying distances and hops, we define a neighborhood space $\mathcal{N}(v)$ for each graph node. The node characteristics of the virtual neighbor nodes within this neighborhood space are identical to those of the actual nodes. The corresponding mathematical formula and its conditions are expressed as follows:

$$\boldsymbol{h}'_j = \boldsymbol{h}_i, \ \forall v'_j \in \mathcal{N}(v_i) \tag{6}$$

$$\begin{cases} v'_j \in \mathcal{N}(v_i), & \text{if } d_{ij} = \underset{i \in \{1,2,\ldots,b\}}{\arg\min}\, d\left(v'_j, v_i\right) \\ v'_j \notin \mathcal{N}(v_i), & \text{others} \end{cases} \tag{7}$$

where v_i and $\boldsymbol{h}_i$ represent the ith actual node and its characteristics, v'_j and $\boldsymbol{h}'_j$ represent the jth virtual node and its characteristics, and $d(v'_j, v_i)$ represents the distance between v'_j and v_i within the positional space $\mathbf{A}_s$.

The deformable region is tasked with selecting virtual neighbor nodes for the central node within the positional space, which results in the loss of the initial graph structural information, while the purpose of graph structure encoding is to embed the local structural information of graph nodes within the nodes themselves. Specifically, the central node and its virtual neighbor nodes reconstitute the local graph structure within the graph structure encoder, as depicted in Fig. 3. To facilitate control over the size of these local subgraphs, the hyperparameter K is employed to define the size of the subgraphs for the central node and its virtual neighbor nodes. Afterward, these subgraphs are processed through an aggregation function that consolidates local structural information, subsequently outputting node features enriched with neighborhood information. The corresponding expression is presented as follows:

$$\tilde{\mathbf{H}}_i^{(l)} = \text{structure encoding}\left(\mathbf{H}_i^{(l)}\right) \tag{8}$$

$$\begin{aligned} \tilde{\boldsymbol{h}}_j &= \mathrm{AGGREGATE}\left(\boldsymbol{h}_q \mid v_q \in \mathcal{N}_j^{(k)}\right) \\ &= \sum_{v_q \in \mathcal{N}_j^{(k)}} \frac{\mathbf{A}_{jq}}{\sqrt{\mathbf{D}_{jj}}\sqrt{\mathbf{D}_{qq}}} \boldsymbol{h}_q, \quad \forall \boldsymbol{h}_j \in \mathbf{H}_i^{(l)} \end{aligned} \tag{9}$$

where $\mathbf{D}$ is the degree matrix of node $\mathbf{A}$, $\mathcal{N}_j^{(k)}$ is the kth-order neighborhood of node j, and AGGREGATE is the aggregation function, which employs the GCN smoothing [54] here.

The node sequences encoded by the graph structure are then projected into Q (query), K (key), and V (value) using three learnable matrices, which can be denoted as

$$\mathbf{Q}_i^{(l)} = \overline{\mathbf{H}}_i^{(l)} \mathbf{W}_q, \mathbf{K}_i^{(l)} = \overline{\mathbf{H}}_i^{(l)} \mathbf{W}_k, \mathbf{V}_i^{(l)} = \overline{\mathbf{H}}_i^{(l)} \mathbf{W}_v \tag{10}$$

where $\mathbf{W}_q \in \mathbb{R}^{d_0 \times d_q}$, $\mathbf{W}_k \in \mathbb{R}^{d_0 \times d_k}$, and $\mathbf{W}_v \in \mathbb{R}^{d_0 \times d_v}$ are learnable parameter matrices. d_0 is the input feature dimension. d_q, d_k, and d_v denote the output dimensions, respectively. The SA scores among different nodes can be expressed as follows:

$$\boldsymbol{\alpha}_i = \frac{\mathbf{Q}_i^{(l)} \mathbf{K}_i^{(l)^T}}{\sqrt{d_q}}. \tag{11}$$

Although the graph structure encoding module assigns local structural information to all graph nodes, this local information alone does not adequately capture the structural relationships between distantly located nodes. To enhance the structural information across the graph, the graph topology encoding is introduced by encoding the relative positional information of nodes into the SA mechanism. For any two nodes, v_i and v_j, their relative distance is defined as the shortest path length between them, that is, the minimum number of edges connecting v_i and v_j, represented by e_{ij}, where e_{ij} is a scalar. By calculating the relative distances between all nodes in the graph, the topology matrix $\mathbf{T}$, which stores the graph's

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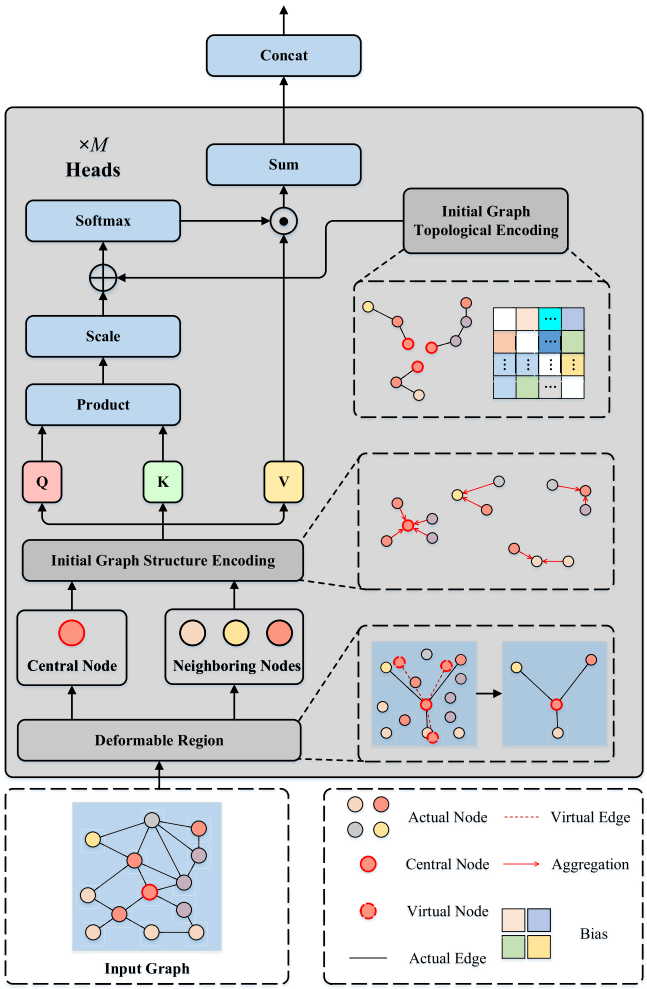


Fig. 3. Structure of the DGSA.

topological structure, is generated. To enable the model to adaptively focus on the virtual neighbor nodes connected to the central node during encoding, graph topology information is incorporated into the attention scores, as detailed in the following expression:

$$\begin{aligned}\boldsymbol{\alpha}_i &= \boldsymbol{\alpha}_i + \text{topology encoding}(\mathbf{T}_i) \\ &= \boldsymbol{\alpha}_i + \boldsymbol{b}_{\mathbf{T}_i(x,y)}\end{aligned} \tag{12}$$

where $\mathbf{T}_i$ is the subtopology matrix corresponding to $\mathbf{H}_i^{(l)}$, $\boldsymbol{b}$ is a learnable parameter, and x and y are used to index the elements in $\mathbf{T}_i$. In summary, the final expression of DGSA is

$$\begin{aligned}&\text{Attention}\left(\mathbf{Q}_i^{(l)}, \mathbf{K}_i^{(l)}, \mathbf{V}_i^{(l)}, \mathbf{T}_i\right) \\ &\quad = \text{Softmax}\left(\frac{\mathbf{Q}_i^{(l)}\mathbf{K}_i^{(l)T}}{\sqrt{d_q}} + b_{\mathbf{T}_i(x,y)}\right)\mathbf{V}_i^{(l)}.\end{aligned} \tag{13}$$

By employing graph structure encoding and graph topology encoding, the relationships between high- and low-order nodes, which are distant from each other in a graph formed from HSI, can be considered within the SA mechanism when modeling global dependencies among different nodes.

D. DGT Layer

The proposed DGT layer comprises two components: the dimensional transformation block and the DGT block. The former consists of a stack of linear layers, batch normalization, and activation functions. It is designed to enhance feature representation capabilities, enabling the transformer encoder to focus on the learning of abstract features across various layers. The corresponding expression is presented as follows:

$$\tilde{\mathbf{H}}^{(l)} = \text{LeakyReLU}\left(\text{BN}\left(\mathbf{H}^{(l)}\mathbf{W}_d^{(l)}\right)\right) \tag{14}$$

where $\mathbf{W}_d^{(l)}$ is the learnable weight matrix of the dimensional transformation block at layer l.

The DGT block is analogous to the original transformer structure, consisting of the multihead DGSA (MHDGSA) and a feed-forward network (FFN). The MHDGSA, which connects and reprojects the outputs of multiple DGSAs, enhances the model's capability to improve the learning of node representations through SA across various subspaces. The mathematical expression for MHDGSA is given as follows:

$$\begin{aligned}&\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}, \mathbf{T}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_M)\mathbf{W}_O^{(l)} \\ &\quad \text{where head}_\text{i} = \text{Attention}\left(\mathbf{Q}^{(l)}, \mathbf{K}^{(l)}, \mathbf{V}^{(l)}, \mathbf{T}\right)\end{aligned} \tag{15}$$

where M is the number of attention heads and $\mathbf{W}_O^{(l)}$ is the output projection matrix of the lth layer MHDGSA. Then, the output of the MHDGSA is combined with the original input through a residual connection and then fed into the FFN. The output from the FFN is subsequently added to this mixture to produce the final output of the DGT layer.

E. Composite Feature Fusion Learning

The graph features extracted by multiple stacked DGT layers encompass both simple low-level and complex high-level characteristics, which are complementary across different forms and layers. For example, shallow DGT layers extract basic low-level features, which provide detailed object boundary information and the immediate structural characteristics of the data. As the network deepens, however, the layers capture more complex and abstract high-level features, thereby enabling a more comprehensive understanding of the relationships between nodes. To fully leverage these features for precise classification of HSI, a CFFL module is proposed. This module adaptively fuses these features, resulting in discriminative representations. The detailed structure is illustrated in Fig. 4.

Initially, features containing diverse information are input into the CFFL module. Given that the graph node features extracted by the DGT layer cannot be directly fused with local pixel-level 3-D cube features, these simple low-level features with complex, abstract high-level features are concatenated and projected into a new feature space. The expression for this projection is given as follows:

$$\mathbf{H}^G = \text{Concat}\left(\mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(L)}\right)\mathbf{W}^G \tag{16}$$

where $\mathbf{W}^G \in \mathbb{R}^{(d_1+d_2+\ldots+d_L)\times d_O}$ is the learnable projection matrix. d_l and d_O denote the output dimension of the DGT at layer l and the output dimension after projection, respectively.

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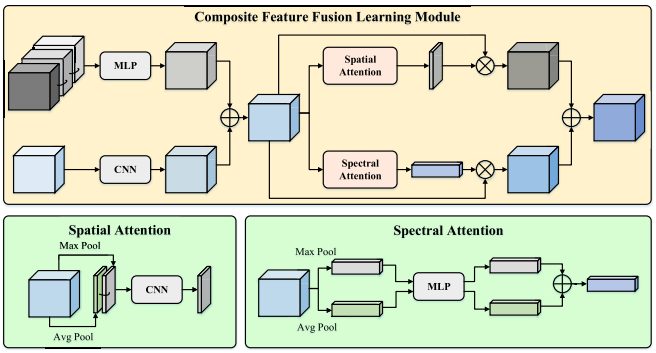


Fig. 4. Schematic of the CFFL.

Furthermore, 3-D pixel-level features are processed through a convolutional layer to enhance the extraction of local features, which are then summed with the decoded graph features for fusion

$$\mathbf{X}^{C}_{h,w,d_O} = \sigma\left(\sum_{i,j,d_0} \mathbf{K}_{i,j,d_0,d_O}\mathbf{X}^{o}_{h,w,d_0} + \mathbf{B}_{i,j,d_0,d_O}\right) \tag{17}$$

$$\mathbf{H}^{C} = \text{Reshape}\left(\mathbf{M}\mathbf{H}^{G}\right) + \mathbf{X}^{C} \tag{18}$$

where $\mathbf{K}$ is a convolution kernel of size Z. The integrated features are subsequently directed into spectral and spatial feature learning modules, with the mathematical expressions detailed in the following:

$$\begin{aligned}\mathbf{H}^{\text{Spe}} &= \mathbf{H}^{C} \odot \sigma\left(\text{MLP}\left(\text{AvgPool}\left(\mathbf{H}^{C}\right) + \text{MaxPool}\left(\mathbf{H}^{C}\right)\right)\right) \\ &= \mathbf{H}^{C} \odot \sigma\left(\mathbf{W}_1\left(\mathbf{W}_0\left(\mathbf{H}^{e}_{\text{avg}} + \mathbf{H}^{e}_{\text{max}}\right)\right)\right) \end{aligned} \tag{19}$$

$$\begin{aligned}\mathbf{H}^{\text{Spa}} &= \mathbf{H}^{C} \odot \sigma\left(\left[\text{AvgPool}\left(\mathbf{H}^{C}\right), \text{MaxPool}\left(\mathbf{H}^{C}\right)\right]\right) * \mathbf{W}_2 \\ &= \mathbf{H}^{C} \odot \sigma\left(\left[\mathbf{H}^{a}_{\text{avg}}, \mathbf{H}^{a}_{\text{max}}\right]\right) * \mathbf{W}_2 \end{aligned} \tag{20}$$

where $\mathbf{W}_0$, $\mathbf{W}_1$, and $\mathbf{W}_2$ are learnable parameters and $\odot$ and $*$ are elementwise multiplication and convolution operations, respectively.

The refined features, having undergone spectral and spatial learning, are combined through summation. These final representations, integrating features from all modalities, are then processed by a fully connected layer. This layer adjusts the number of feature maps and uses the softmax function to normalize the outputs, producing probabilities for different categories, which can be expressed as

$$\mathbf{Y} = \text{Softmax}\left(\mathbf{W}^{F}\left(\mathbf{H}^{\text{Spa}} + \mathbf{H}^{\text{Spe}}\right) + \mathbf{B}^{F}\right) \tag{21}$$

where $\mathbf{W}^{F}$ and $\mathbf{B}^{F}$ denote learnable parameters and bias, respectively. Based on these probabilities, we utilize a cross-entropy function to calculate the loss function and employ backpropagation to optimize the proposed DGT model.

III. Experiments

In this article, four datasets named Indian Pines, Salinas, WHU-Hi-HongHu, and Yellow River Estuary Coastal Wetland are utilized to assess the classification performance of the proposed DGT. A random sampling method is used to partition these datasets into training, validation, and testing sets, with training ratios set at 3%, 0.3%, 0.3%, and 0.1%, respectively.

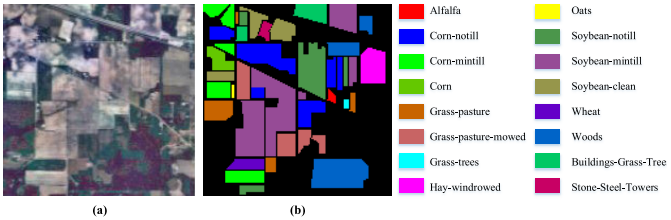


Fig. 5. (a) False-color composition of the Indian Pines dataset. (b) Ground truth-map containing 16 land-cover classes.

TABLE I
Number of Training, Validation, and Testing Samples for the Indian Pines Dataset

No.	Class	Train	Val	Test	Total
1	Alfalfa	2	2	42	46
2	Corn-notill	43	43	1342	1428
3	Corn-mintill	25	25	780	830
4	Corn	8	8	221	237
5	Grass-pasture	15	15	453	483
6	Grass-trees	22	22	686	730
7	Grass-pasture-mowed	1	1	26	28
8	Hay-windrowed	15	15	448	478
9	Oats	1	1	18	20
10	Soybean-notill	30	30	912	972
11	Soybean-mintill	74	74	2307	2455
12	Soybean-clean	18	18	557	593
13	Wheat	7	7	191	205
14	Woods	38	38	1189	1265
15	Buildings-Grass-Trees	12	12	362	386
16	Stone-Steel-Towers	3	3	87	93
	Total	314	314	9621	10249

A. Datasets

1) Indian Pines Dataset: The Indian Pines dataset was collected using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Northwestern Indiana, USA. It consists of 145 × 145 pixels, each with a spatial resolution of 20 × 20 m. It retains 200 spectral bands after the removal of 20 water-absorbing bands. Besides the background, the dataset includes 16 distinct land cover classes, which can help evaluate the model's classification results. Table I displays the division of each category within the dataset, and Fig. 5 illustrates the false-color image, the truth map, and the corresponding color table.

2) Salinas Dataset: The Salinas dataset was acquired using the AVIRIS imaging spectrometer, capturing the spatial and spectral information of Salinas Valley, CA, USA. This dataset comprises a spatial configuration of 512 × 217 pixels, with a spatial resolution of 3.7 × 3.7 m. Similar to the Indian Pines dataset, after the removal of 20 water-absorbing bands, 204 spectral bands remain. In this dataset, a total of 54 129 samples have been categorized into 16 classes of land cover types. Table II details the distribution of training, validation, and testing sets across these classes. Fig. 6 illustrates the false-color image, the truth map, and the corresponding color table.

3) WHU-Hi-HongHu Dataset: This dataset was collected in Honghu, Hubei, China, using a 17-mm focal length Headwall Nano-Hyperspec imaging sensor mounted on

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TABLE II
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE SALINAS DATASET

No.	Class	Train	Val	Test	Total
1	Brocoli_green_weeds_1	7	7	1995	2009
2	Brocoli_green_weeds_2	12	12	3702	3726
3	Fallow	6	6	1964	1976
4	Fallow_rough_plow	5	5	1384	1394
5	Fallow_smooth	9	9	2660	2678
6	Stubble	12	12	3935	3959
7	Celery	11	11	3557	3579
8	Grapes_untrained	34	34	11203	11271
9	Soil_vinyard_develop	19	19	6165	6203
10	Corn_senesced_green_weeds	10	10	3258	3278
11	Lettuce_romaine_4wk	4	4	1060	1068
12	Lettuce_romaine_5wk	6	6	1915	1927
13	Lettuce_romaine_6wk	3	3	910	916
14	Lettuce_romaine_7wk	4	4	1062	1070
15	Vinyard_untrained	22	22	7224	7268
16	Vinyard_vertical_trellis	6	6	1795	1807
	Total	170	170	53789	54129

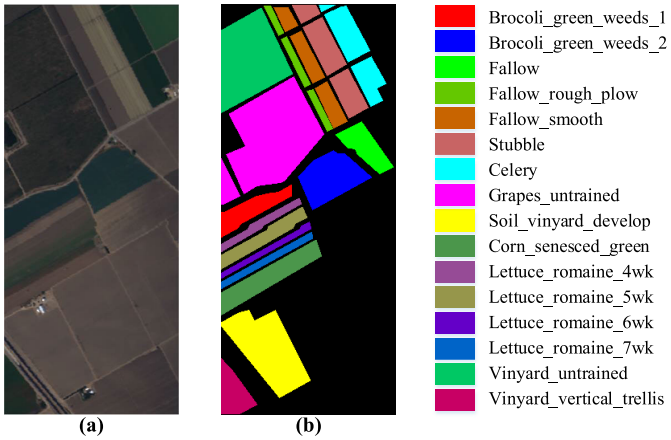


Fig. 6. (a) False-color composition of the Salinas dataset. (b) Ground truth-map containing 16 land-cover classes.

TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE HONGHU DATASET

No.	Class	Train	Val	Test	Total
1	Red roof	43	43	13955	14041
2	Road	11	11	3490	3512
3	Bare soil	66	66	21689	21821
4	Cotton	490	490	162305	163285
5	Cotton firewood	19	19	6180	6218
6	Rape	134	134	44289	44557
7	Chinese cabbage	73	73	23957	24103
8	Pakchoi	13	13	4028	4054
9	Cabbage	33	33	10753	10819
10	Tuber mustard	38	38	12318	12394
11	Brassica parachinensis	34	34	10947	11015
12	Brassica chinensis	27	27	8900	8954
13	Small Brassica chinensis	68	68	22371	22507
14	Lactuca sativa	23	23	7310	7356
15	Celtuce	4	4	994	1002
16	Film covered lettuce	22	22	7218	7262
17	Romaine lettuce	10	10	2990	3010
18	Carrot	10	10	3197	3217
19	White radish	27	27	8658	8712
20	Garlic sprout	11	11	3464	3486
21	Broad bean	4	4	1320	1328
22	Tree	13	13	4014	4040
	Total	1173	1173	384347	386693

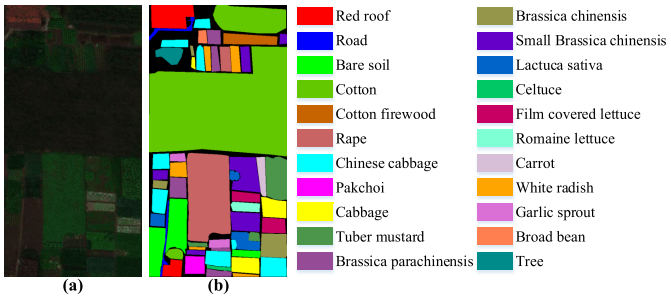


Fig. 7. (a) False-color composition of the HongHu dataset. (b) Ground truth-map containing 16 land-cover classes.

a DJI Matrice 600 Pro UAV platform. The imagery comprises 940 × 475 pixels and encompasses 270 spectral bands. In this dataset, 22 land cover types have been identified and labeled. The distribution of samples across these categories is detailed in Table III. Fig. 7 illustrates the false-color image, the truth map, and the corresponding color table.

4) Yellow River Estuary Coastal Wetland Dataset: The YRE dataset was captured by the Gaofen-5 satellite over the coastal wetland at the mouth of the Yellow River in Bohai Bay and Laizhou Bay. This dataset contains 296 spectral bands and 740 × 761 pixels across eight different land cover categories. For details of each category and the dataset division, refer to Table IV. Fig. 8 illustrates the false-color image, the truth map, and the corresponding color table.

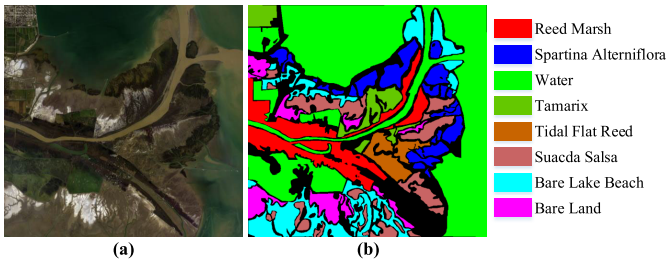


Fig. 8. (a) False-color composition of the YRE dataset. (b) Ground truth-map containing 16 land-cover classes.

B. Experimental Settings

1) Comparison Algorithms: In the experiments, to demonstrate the performance of the proposed DGT, nine state-of-the-art (SOTA) deep learning methods are utilized for comparative analysis in both quantitative and qualitative classification outcomes. These include CNN-based methods such as the 3DCNN [24], deep feature fusion network (DFFN) [30], and residual spectral–spatial attention network (RSSAN) [31]; graph transformer-based methods such as graph transformer with all nodes (GT-all) and graph transformer with adjacency nodes (GT-adj) [55], which are aimed at aggregating features from higher order nodes; and transformer-based methods, specifically four different classification approaches named SpectralFormer (SF) [36], spectral–spatial feature tokenization transformer (SSFTT) [37], GAHT network [39], and GraphGST [41]. All the aforementioned algorithms employ

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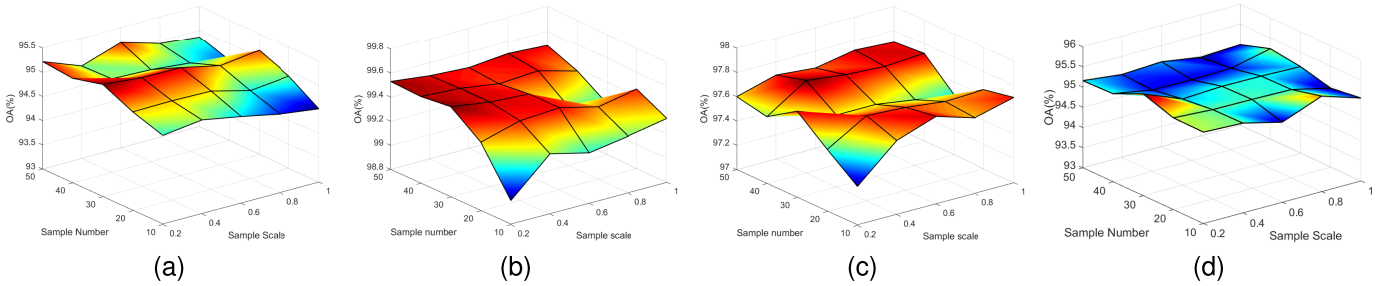


Fig. 9. Classification performance of each method with different sample scales and numbers on four datasets. (a) Indian Pines. (b) Salinas. (c) WHU-Hi-Honghu. (d) YRE.

TABLE IV
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE YRE DATASET

No.	Class	Train	Val	Test	Total
1	Reed Marsh	45	45	44349	44439
2	Spartina Alterniflora	31	31	30656	30718
3	Water	215	215	213979	214409
4	Tamarix	17	17	16277	16311
5	Tidal Flat Reed	13	13	12220	12246
6	Suacda Salsa	30	30	29529	29589
7	Bare Lake Beach	47	47	45997	46091
8	Bare Land	22	22	21254	21298
	Total	420	420	414261	415101

TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT SUBGRAPH SIZES ON EACH DATASET

Dataset	K	1	2	3	4	5
IP	OA	**95.32**	95.11	95.22	95.04	95.14
	AA	**95.08**	94.60	94.86	94.45	94.66
	Kappa	**94.66**	94.42	94.55	94.35	94.45
SA	OA	**99.55**	99.46	99.33	99.35	99.25
	AA	**99.32**	99.18	98.99	99.18	98.92
	Kappa	**99.50**	99.40	99.25	99.27	99.17
HH	OA	**97.71**	97.61	97.58	97.60	97.54
	AA	**95.13**	94.88	94.94	94.86	94.72
	Kappa	**97.11**	96.98	96.95	96.97	96.89
YRE	OA	95.08	95.10	95.45	95.19	**95.52**
	AA	91.45	91.66	92.08	91.51	**92.09**
	Kappa	92.90	92.94	93.44	93.07	**93.54**

parameter settings consistent with their published versions to ensure optimal classification performance. In addition, to reduce the effects of randomness and ensure stability in the experimental results, the outcomes from all methods are averaged over ten independent runs, and the corresponding standard deviations are provided.

2) Evaluation Metrics: Four evaluation metrics, namely, class accuracy (CA), overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ), are employed to assess the classification performance of the proposed DGT. In addition, these metrics are utilized for quantitative analyses in comparative experiments.

3) Implementation Details: The proposed DGT method is implemented using the deep learning library PyTorch, with Python and PyTorch versions specified as 3.9 and 1.12, respectively. The Adam optimizer is utilized to efficiently update the model parameters, aiming to minimize the loss function during training. The learning rate is set at 0.0005, with the training epochs configured to 300. All experiments are conducted on a server equipped with an NVIDIA GeForce RTX 3090 graphics processing unit (GPU).

C. Parameter Sensitivity Analysis

1) Influence of Sampled Number N and Scale λ: The hyperparameters N and λ are crucial in the DGSA, dictating the density and extent of the sampled virtual neighbor nodes, respectively. A larger λ results in a greater maximum distance between the virtual nodes' offset positions and the central node. To explore the impact of virtual neighbor nodes from varying distances and quantities on the DGT, different combinations of the parameters (λ and N) are employed in experiments, with the results depicted in Fig. 9. It is observed that as the number of sampled virtual neighbor nodes increases from 10 to 30, the OA of the DGT on the four datasets generally rises and then stabilizes. This enhancement is attributed to the model's enhanced capability to discern differences between classes as the number of sampled nodes increases. The effect of the sampling scale is contingent upon the number of sampled nodes, and with fewer nodes, the classification results improve as the sampling scale increases. Conversely, with a larger number of nodes, a smaller sampling scale typically yields better performance, illustrating the complementary relationship between the number of samples and the sampling scale. For consistency across the four datasets, the parameter combination used is (0.2, 30).

2) Influence of Graph Structure Size K: The hyperparameter K determines the size of the local subgraph structure that the input node sequence can capture within the DGSA. To explore the impact of K on the classification performance of the proposed DGT, experiments are conducted across four datasets. The values of K are set at 1, 2, 3, 4, and 5, and the results for OA, AA, and κ are displayed in Table V. On the YRE dataset, classification performance peaks when the subgraph size, K, is set to 5, whereas, on the other three datasets, the optimal performance occurs at $K = 1$. This variation can be attributed to the increase in the subgraph size allowing more diverse class node information to converge on the central node. Specifically, the YRE dataset features predominantly large homogeneous regions, resulting in subgraphs with a higher concentration of similar class nodes. To ensure

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TABLE VI
CLASSIFICATION ACCURACY FOR DIFFERENT LAYERS ON EACH DATASET

Dataset	L	1	2	3	4	5
IP	OA	94.49	94.59	**95.32**	95.00	94.94
	AA	93.58	94.25	**95.08**	94.90	95.07
	Kappa	93.71	93.83	**94.66**	94.30	94.22
SA	OA	98.76	99.18	**99.55**	99.50	99.51
	AA	98.89	98.76	**99.32**	99.24	99.21
	Kappa	98.63	99.09	**99.50**	99.44	99.46
HH	OA	97.59	97.62	**97.71**	97.48	97.46
	AA	94.72	94.83	**95.13**	94.66	94.77
	Kappa	96.95	96.99	**97.11**	96.82	96.79
YRE	OA	95.22	95.44	**95.52**	95.48	95.42
	AA	91.42	91.60	92.09	**92.26**	91.67
	Kappa	93.11	93.43	**93.54**	93.49	93.39

TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT KERNEL SIZES ON EACH DATASET

Dataset	Z	3	5	7	9	11
IP	OA	94.25	95.03	**95.32**	95.06	95.11
	AA	93.92	94.43	**95.08**	94.60	94.56
	Kappa	93.44	94.33	**94.66**	94.37	94.42
SA	OA	99.11	99.05	99.35	99.51	**99.55**
	AA	98.74	98.56	98.99	99.31	**99.32**
	Kappa	99.01	98.94	99.28	99.46	**99.50**
HH	OA	97.22	97.37	97.57	97.62	**97.71**
	AA	93.97	94.45	94.75	94.75	**95.13**
	Kappa	96.48	96.68	96.92	96.99	**97.11**
YRE	OA	94.61	94.89	95.15	95.27	**95.52**
	AA	90.72	90.92	91.68	91.80	**92.09**
	Kappa	92.23	92.63	93.00	93.18	**93.54**

the best performance, $K = 5$ is set for the YRE dataset and $K = 1$ is set for the other three datasets.

3) Influence of Network Depth L: In general, increasing the number of transformer encoders enhances the model's representational and feature extraction capabilities. However, adding more encoders also raises the model's complexity and computational costs, potentially leading to overfitting due to its substantial capability for fitting complex functions. For GNNs, increasing the network depth may lead to an oversmoothing issue. This occurs because the aggregation of high-order neighbor node information by the model makes it challenging to distinguish central node features from those of other node classes. To thoroughly investigate the impact of encoder quantity on the proposed DGT, experiments are conducted across four different datasets using DGTs with varying numbers of layers. The results are presented in Table VI. The performance of shallow networks is inferior to that of deeper networks due to the limited feature representation capabilities of shallow models. As the depth of the network increases to a certain level, performance begins to plateau, which indicates that DGT possesses considerable robustness. Considering the balance between network depth and classification performance, a configuration of three DGT layers is selected for the experiments.

4) Influence of Kernel Size Z: Z is a crucial hyperparameter within the CFFL module, governing the size of the local receptive field and significantly influencing the module's ability to capture local detailed features. A larger Z allows the CFFL to learn local boundary structure information from HSI, but it also increases the number of learnable parameters to some extent. With all other hyperparameters held constant, the value of Z is selected from the set 3, 5, 7, 9, 11. Table VII illustrates the classification performance of the DGT under varying settings of the hyperparameter Z. On the Indian Pines dataset, classification performance initially improves with the increase of convolution kernel size but declines after peaking at $Z = 7$. Conversely, for the remaining three datasets, the optimal performance is observed at $Z = 11$. This distinction is attributed to the larger, smoother regions in these datasets compared to Indian Pines, which are more amenable to processing with larger convolution kernels. Consequently, the selected sizes of the convolution kernels for the Indian Pines, Salinas, HongHu, and YRE datasets are 7, 11, 11, and 11, respectively.

D. Classification Performance

In this section, nine different HSI classification algorithms—3DCNN, DFFN, RSSAN, GT-all, GT-adj, SF, SSFTT, GAHT, and GraphGST—are employed to assess the classification performance of the proposed DGT. A series of experiments are conducted for evaluation on four datasets: Indian Pines, Salinas, HongHu, and YRE.

Tables VIII–XI display the results for all methods, including CA, OA, AA, κ, and their standard deviations after ten independent experiments, with the best results highlighted in bold. On the Indian Pines dataset, the proposed DGT surpasses other methods, achieving OA, AA, and κ values of 95.32%, 95.08%, and 94.66%, respectively. For the Salinas dataset, these three metrics for the DGT exceed 99%. In addition, the DGT's classification outcomes also surpass those of comparative algorithms on the HongHu and YRE datasets. It is noted that comparative algorithms perform poorly on the Indian Pines dataset, which is due to the extreme imbalance of the samples in this dataset. The classification performance of GT-all is worse than that of GT-adj, which reflects that selecting all graph nodes as keys is more prone to causing oversmoothing issues. 3DCNN, DFFN, and RSSAN perform better on the HongHu and YRE datasets, which have larger training samples, yet their performance still falls short of transformer-based methods, which proves that global information is one of the most important factors in HSI classification. GAHT achieves results comparable to GraphGST on the Salinas, HongHu, and YRE datasets but fares poorly on the Indian Pines dataset, reflecting GraphGST's superior robustness. The DGT integrates the global modeling capabilities of transformers with the graph's strength in processing irregular data, which can consider both global and local information effectively. The experimental results also confirm the efficacy of the DGT.

In addition, classification maps from various methods are compared for qualitative analysis. Figs. 10–13 depict the classification maps derived from the Indian Pines, Salinas, HongHu, and YRE datasets, respectively. On the Indian Pines dataset, methods such as 3DCNN and RSSAN exhibit numerous misclassifications due to the dataset's complex feature distribution and these methods' emphasis on localized spatial–spectral information. Conversely, the remaining three

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TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE INDIAN PINES DATASET

Class	3DCNN	DFFN	RSSAN	GT-all	GT-adj	SF	SSFTT	GAHT	GraphGST	DGT
1	1.82±3.64	28.64±13.61	14.55±10.42	71.97±35.33	82.94±26.51	18.18±6.27	40.00±20.61	30.00±8.55	**99.55±0.91**	94.36±3.47
2	53.76±4.52	82.56±3.13	78.36±2.16	88.46±4.84	86.33±6.43	64.35±3.97	80.15±5.24	86.36±3.49	**94.01±2.16**	93.49±2.97
3	40.15±13.26	79.67±11.18	66.81±8.00	83.96±10.25	88.5±5.56	53.09±7.22	73.47±6.06	84.10±7.43	91.93±1.97	**92.59±3.33**
4	17.04±7.14	71.12±7.70	31.66±2.62	77.11±9.4	86.92±8.7	46.91±8.94	74.35±7.72	83.50±6.94	85.02±8.82	**95.86±2.96**
5	53.19±8.93	89.58±2.67	71.65±24.40	86.83±4.09	88.46±6.33	64.13±6.87	87.78±3.15	87.47±3.77	90.88±4.1	**91.92±5.47**
6	95.14±3.99	96.16±1.96	93.68±3.52	96.01±1.79	96.35±2.26	89.37±4.69	96.71±1.25	98.40±0.93	**99.58±0.3**	97.56±1.16
7	0.77±1.54	56.15±18.30	11.54±8.43	98.49±3.02	69.97±44.29	41.54±20.41	63.85±16.78	86.92±13.67	95.19±4.7	**99.63±1.11**
8	98.00±2.19	99.42±0.41	96.36±4.08	94.62±6.18	98.15±3.98	97.02±2.07	98.76±1.42	99.96±0.09	**100.0±0.0**	99.93±0.14
9	14.44±11.97	33.33±11.65	17.78±13.33	72.26±34.5	72.2±34.47	35.56±7.54	85.56±8.31	48.89±12.86	**91.05±14.52**	88.98±26.18
10	9.29±4.02	83.54±5.20	74.75±5.79	77.69±8.96	81.86±8.05	69.03±3.61	81.05±6.57	86.27±5.72	**93.14±4.44**	90.54±5.44
11	81.96±4.41	85.88±2.93	83.11±3.58	89.93±3.15	94.46±2.51	79.43±2.41	86.45±3.89	89.95±3.72	94.2±2.14	**96.95±1.67**
12	25.16±3.67	74.05±2.84	64.12±7.89	83.39±8.86	85.39±7.51	39.43±4.62	77.92±7.35	84.41±5.13	86.75±5.17	**90.89±5.17**
13	96.99±3.27	96.37±6.04	93.37±6.07	95.47±4.57	99.06±0.72	96.48±2.50	**99.48±0.66**	98.86±1.24	99.09±0.44	99.27±0.57
14	92.61±2.70	96.15±2.39	95.56±1.28	95.19±3.5	96.88±3.71	91.13±4.44	95.39±1.63	95.80±2.24	98.49±2.58	**99.16±1.27**
15	57.25±2.68	77.30±4.15	52.12±17.78	84.36±12.22	94.18±7.18	60.28±9.77	83.69±5.18	79.89±7.82	92.94±5.59	**96.27±6.65**
16	95.40±1.26	95.86±3.05	78.16±13.77	92.08±8.88	91.36±9.93	**99.54±0.92**	95.17±2.45	97.47±2.85	98.0±2.47	93.81±6.35
OA(%)	67.18±1.14	86.01±1.71	78.78±2.13	88.39±2.45	91.16±2.14	72.73±1.45	85.67±1.22	89.34±1.58	94.38±0.8	**95.32±0.9**
AA(%)	54.56±1.40	77.86±2.90	63.97±3.21	86.74±4.38	88.31±3.56	65.34±2.82	82.49±1.73	83.64±1.09	94.36±1.15	**95.08±1.36**
Kappa(%)	61.86±1.37	84.07±1.97	75.73±2.46	86.76±2.84	89.92±2.44	68.84±1.73	83.67±1.37	87.84±1.79	93.59±0.92	**94.66±1.03**

TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE SALINAS DATASET

Class	3DCNN	DFFN	RSSAN	GT-all	GT-adj	SF	SSFTT	GAHT	GraphGST	DGT
1	77.47±18.57	94.92±1.87	91.30±7.41	86.16±30.01	89.17±12.39	95.53±0.69	98.66±1.64	97.68±3.39	97.63±3.76	**99.99±0.03**
2	99.51±0.28	99.15±0.35	99.49±0.35	99.03±1.85	98.91±2.26	98.40±0.72	99.84±0.10	99.98±0.02	99.94±0.1	**100.0±0.0**
3	69.89±9.82	84.24±8.91	63.55±33.80	94.16±8.92	100.0±0.0	78.31±12.00	61.58±25.24	90.37±15.73	91.65±8.14	**100.0±0.0**
4	98.95±0.76	97.13±1.07	79.26±39.63	98.2±2.36	97.6±3.06	96.05±0.71	99.28±0.96	99.21±0.35	98.49±1.54	**99.78±0.28**
5	96.04±1.18	92.09±3.14	95.51±3.50	97.48±0.65	97.24±1.1	82.40±2.38	97.19±1.54	97.17±2.91	**98.66±0.97**	98.09±0.8
6	99.08±0.63	99.12±0.58	99.59±0.42	99.06±1.56	99.84±0.17	99.13±0.88	99.23±1.37	99.76±0.30	99.86±0.04	**99.88±0.13**
7	99.20±0.33	98.44±0.73	95.91±2.20	98.8±0.71	98.76±2.4	97.02±1.03	96.73±1.70	98.11±2.16	98.98±0.37	**99.98±0.03**
8	78.75±3.56	83.04±2.24	75.76±9.01	90.74±4.61	93.7±5.81	77.32±3.74	84.39±6.71	85.67±3.77	83.04±6.56	**99.68±0.4**
9	97.76±0.87	95.58±1.04	98.79±0.82	99.43±1.52	99.78±0.34	94.30±1.74	97.85±1.67	98.86±0.84	99.66±0.59	**99.91±0.27**
10	87.03±3.04	87.88±5.16	74.45±36.21	92.58±2.26	95.22±1.99	81.32±6.69	90.86±3.20	93.24±2.35	92.8±2.05	**98.87±1.01**
11	40.85±22.37	86.33±7.95	48.36±38.96	90.12±3.55	90.28±9.26	71.60±20.73	88.00±8.21	95.24±4.13	93.34±4.88	**99.69±0.64**
12	99.04±1.59	99.56±0.35	78.75±39.39	98.37±3.38	99.55±0.66	96.09±4.06	99.34±0.50	99.49±0.90	97.29±3.34	**99.63±0.53**
13	94.66±2.37	94.62±4.00	57.60±47.08	93.36±10.34	87.48±23.24	87.65±8.37	92.92±5.37	76.62±15.23	96.33±5.17	**98.47±1.8**
14	97.20±0.89	96.88±2.97	70.66±36.58	94.62±4.73	88.81±11.82	96.13±2.18	94.17±4.12	**98.87±0.73**	96.2±3.37	95.73±3.49
15	68.11±5.15	76.13±4.21	81.06±8.32	80.99±28.08	86.8±17.02	77.92±5.51	70.93±17.28	84.23±3.83	86.13±6.93	**99.52±0.49**
16	79.27±6.37	78.05±7.70	90.35±6.18	91.16±9.03	92.14±16.12	84.76±4.93	92.95±4.55	93.14±1.83	95.1±4.52	**99.94±0.17**
OA(%)	85.90±1.36	89.40±1.06	84.58±6.94	93.12±3.24	94.93±2.98	86.73±0.59	89.42±2.08	92.86±1.30	93.02±1.08	**99.55±0.13**
AA(%)	86.42±2.59	91.45±1.45	81.27±13.49	94.02±2.69	94.7±1.77	88.37±1.25	91.50±2.50	94.23±1.99	95.32±0.94	**99.32±0.26**
Kappa(%)	84.28±1.52	88.20±1.19	82.80±7.75	92.33±3.65	94.34±3.34	85.26±0.65	88.21±2.37	92.05±1.44	92.25±1.2	**99.5±0.15**

datasets contain large homogeneous areas, and classification maps obtained from patch-based transformer methods such as SF, SSFTT, and GAHT are characterized by noticeable noise, which reflects the importance of interpatch dependencies. Ultimately, it is evident across all four datasets that the proposed DGT maintains intricate boundary details and produces smoother classification maps, which are closer to the ground truth.

IV. DISCUSSION

A. Ablation Study

The proposed DGT comprises multiple components, including the LAPE module, graph structure encoding, graph topology encoding, deformable transformer encoder, and CFFL module. To assess the necessity and effectiveness of these distinct modules within the DGT, five variants formed by the absence of these components are utilized to conduct ablation studies across four datasets. In addition, the benchmark is added, which is a network consisting of a two-layer graph transformer encoder [55]. The results are presented in Fig. 14.

1) Learnable Absolute Position Encoding: This experiment validates the contribution of the proposed LAPE to DGT. Since this module is fundamental for generating virtual neighbor nodes within the DGT, it cannot be directly omitted. Instead, a degraded version of the module is utilized, where all learnable parameters are removed. The specific results are depicted in Fig. 14. On all four datasets, the classification performance is lower than that of the complete DGT. This decline is attributed to the absence of node position embedding, which impairs the model's ability to discern positional dependencies among elements in the input node sequence. This outcome further underscores the efficacy of the designed LAPE.

2) DGT Layer: Serving as a crucial backbone network in the proposed method, DGT plays a pivotal role in the feature extraction task. As can be seen from the results, after losing the DGT, the network's OA, AA, and κ on the four datasets drop dramatically, and its performance is even lower than that of the benchmark. These significant reductions underscore the essential and irreplaceable role of the proposed DGT within the entire model.

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TABLE X
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE WHU-HI-HONGHU DATASET

Class	3DCNN	DFFN	RSSAN	GT-all	GT-adj	SF	SSFTT	GAHT	GraphGST	DGT
1	90.92±4.06	94.48±1.09	93.59±1.74	96.58±1.87	97.06±3.01	92.92±2.01	94.82±1.84	95.11±0.49	96.63±1.83	**98.64±0.4**
2	78.18±5.14	73.27±8.40	79.75±7.16	73.8±8.33	75.56±8.82	75.99±8.15	78.74±7.22	82.79±3.94	70.03±9.94	**84.95±7.82**
3	90.33±1.78	90.69±2.51	91.65±3.97	94.44±2.23	96.63±1.54	87.15±3.40	89.72±2.96	89.48±3.76	94.45±1.59	**97.9±1.27**
4	99.35±0.06	99.28±0.21	99.27±0.15	99.37±0.34	99.52±0.16	99.10±0.18	99.31±0.12	99.48±0.21	99.35±0.29	**99.67±0.12**
5	49.59±8.42	67.24±3.82	72.06±7.97	83.86±5.66	89.82±3.84	62.93±7.89	64.96±10.26	78.23±4.73	82.96±5.04	**94.06±2.58**
6	93.70±1.67	95.29±0.19	94.01±1.36	96.75±1.32	98.01±0.48	94.21±0.66	96.70±0.98	97.53±0.74	97.05±1.17	**99.06±0.33**
7	78.90±1.70	80.22±3.30	81.13±1.99	89.53±2.18	93.76±1.27	79.69±3.26	80.14±2.88	87.67±2.89	87.28±5.26	**95.96±1.43**
8	12.04±5.50	23.76±11.13	24.42±4.75	54.2±11.19	78.43±12.13	20.50±5.25	29.19±5.68	38.07±10.13	53.54±11.52	**95.89±4.12**
9	95.99±0.44	96.24±2.13	96.25±1.18	98.06±0.99	97.69±1.66	97.64±1.07	96.82±2.03	98.34±1.39	97.21±1.8	**98.54±1.65**
10	71.35±5.17	76.37±6.03	71.96±3.46	89.46±2.96	**93.5±3.02**	60.04±3.55	83.14±3.02	88.58±2.29	83.06±5.19	93.48±2.86
11	60.31±2.20	63.98±3.64	69.84±6.26	87.5±3.26	90.08±3.65	66.08±2.23	67.85±4.07	82.19±2.46	84.79±6.96	**95.79±1.03**
12	62.47±3.31	65.62±4.07	66.29±3.88	75.08±10.74	89.01±4.34	59.33±2.65	65.33±1.53	74.02±6.88	53.86±18.84	**92.68±3.08**
13	78.89±2.82	79.11±1.31	79.06±2.83	86.46±2.89	90.59±2.63	76.54±4.26	77.67±3.69	83.52±3.23	79.73±6.97	**95.98±1.74**
14	75.74±6.88	88.35±1.67	75.98±2.16	87.42±7.89	91.81±8.0	82.19±0.91	83.70±5.16	91.38±5.00	81.54±9.47	**95.13±1.44**
15	36.12±6.75	48.84±21.12	57.03±22.70	85.11±9.77	81.31±14.04	50.02±15.09	49.38±12.47	74.38±11.34	89.99±3.12	**93.11±5.67**
16	87.85±4.80	89.34±3.37	83.88±6.03	94.64±2.16	97.45±2.04	88.68±2.13	88.62±3.54	89.25±7.11	93.01±2.86	**97.92±1.49**
17	55.25±11.70	79.61±10.94	76.15±7.63	79.19±14.75	91.77±5.46	74.48±17.39	82.32±4.32	86.24±11.88	87.56±5.27	**94.13±2.86**
18	50.63±5.31	72.44±11.48	65.79±9.20	82.33±8.57	84.42±5.4	65.03±7.73	67.58±11.01	80.44±3.33	90.11±7.69	**92.35±3.57**
19	80.86±2.92	80.94±5.27	84.40±3.54	92.26±2.46	89.9±5.1	81.99±2.15	87.71±2.49	90.32±2.75	**93.48±4.19**	91.05±2.72
20	71.35±4.45	89.56±2.88	81.56±7.10	85.62±9.17	77.25±10.45	87.66±4.10	81.86±4.97	86.66±6.88	85.53±6.22	**92.44±6.33**
21	0.95±1.69	3.98±3.09	13.03±5.33	57.27±17.35	75.21±19.5	17.03±11.59	36.59±15.36	42.71±6.72	72.11±8.95	**97.34±1.89**
22	55.81±4.82	71.85±5.77	66.64±5.65	86.4±6.84	91.88±5.82	65.95±8.81	69.36±5.76	75.21±7.21	82.09±7.31	**96.82±3.29**
OA(%)	87.51±0.37	89.63±0.90	89.33±0.70	93.9±0.7	95.86±0.49	88.11±0.45	90.14±0.26	92.82±0.21	92.58±0.84	**97.71±0.31**
AA(%)	67.12±0.71	74.11±2.60	73.81±2.08	85.24±2.28	89.58±1.92	72.05±1.46	75.98±0.86	82.35±1.00	84.33±1.63	**95.13±0.57**
Kappa(%)	84.11±0.47	86.87±1.14	86.48±0.89	92.28±0.89	94.77±0.61	84.94±0.57	87.49±0.34	90.91±0.27	90.61±1.07	**97.11±0.39**

TABLE XI
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE YRE DATASET

Class	3DCNN	DFFN	RSSAN	GT-all	GT-adj	SF	SSFTT	GAHT	GraphGST	DGT
1	89.66±5.13	91.26±1.40	90.82±0.91	90.41±2.3	91.33±2.13	89.54±0.94	91.13±1.61	93.37±1.78	90.74±2.86	**93.49±1.43**
2	94.27 ±1.80	93.28 ±1.83	93.12 ±1.18	95.15±1.9	95.52±1.71	94.63 ±1.63	93.69 ±1.48	94.50 ±1.64	95.65±0.85	**95.72±1.43**
3	99.05 ±0.13	98.85 ±0.28	99.00 ±0.22	99.04±0.38	99.11±0.33	98.90 ±0.16	99.08 ±0.18	99.29 ±0.20	98.84±0.26	**99.46±0.38**
4	72.58 ±7.68	72.47 ±3.42	74.73 ±4.75	71.36±12.76	84.81±7.35	73.93 ±6.97	76.81 ±4.99	74.55 ±4.93	72.48±5.16	**90.8±3.61**
5	70.70 ±3.97	61.30 ±15.52	73.64 ±6.64	72.82±5.8	81.16±6.75	77.40 ±8.05	78.77 ±5.82	82.41 ±5.14	78.78±4.0	**89.39±5.67**
6	81.00 ±1.73	78.61 ±2.96	78.86 ±4.92	77.29±5.76	78.26±9.15	75.94 ±6.42	75.07 ±2.59	78.70 ±2.34	77.7±2.99	**81.53±6.12**
7	87.42 ±1.28	89.65 ±2.11	87.21 ±2.46	86.7±4.84	84.04±3.35	88.26 ±2.19	81.54 ±4.13	88.87 ±2.54	87.06±1.85	**91.68±2.88**
8	91.99±1.92	93.15 ±1.55	92.00 ±1.24	90.57±4.69	93.0±4.41	93.99 ±1.59	91.23 ±5.37	**94.79 ±3.58**	89.59±1.04	94.63±2.33
OA(%)	92.88±0.27	92.72±0.64	92.89±0.28	92.61±0.68	93.45±0.54	92.89±0.25	92.29±0.57	93.98±0.23	92.82±0.32	**95.52±0.55**
AA(%)	85.83 ±0.65	84.82 ±2.07	86.17 ±0.96	85.42±1.89	88.4±1.31	86.57 ±0.68	85.91 ±0.90	88.31 ±1.14	86.36 ±0.75	**92.09±0.75**
Kappa(%)	89.71 ±0.39	89.51 ±0.92	89.74 ±0.40	89.34±0.99	90.55±0.79	89.75 ±0.36	88.89 ±0.82	91.30 ±0.34	89.66 ±0.45	**93.54±0.78**

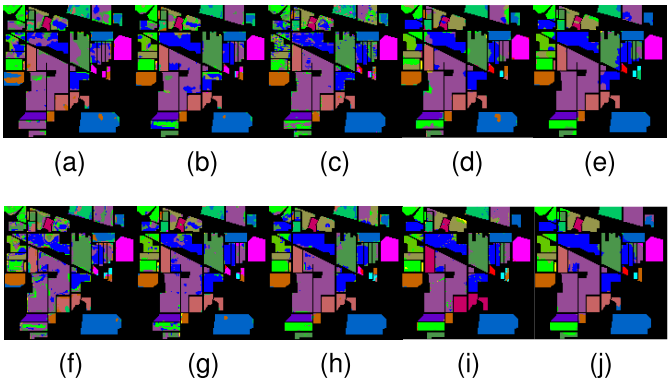


Fig. 10. Classification maps obtained by different methods on the Indian Pines dataset. (a) 3DCNN. (b) DFFN. (c) RSSAN. (d) GT-all. (e) GT-adj. (f) SF. (g) SSFTT. (h) GAHT. (i) GraphGST. (j) DGT.

3) Graph Structure Encoding and Graph Topology Encoding: To further explore the significance of graph structure encoding and graph topology encoding within the proposed DGT, these two modules are individually removed and tested across four datasets. The classification results are elaborately depicted in Fig. 14. It is observed that the classification performance of the model incorporating graph structure encoding surpasses that of the model with graph topology encoding, yet both show a decline compared to the complete DGT. This disparity arises because the graph structure encoding includes neighborhood structural information of nodes, while graph topology encoding captures the long-distance topological relationships between nodes. These elements are mutually complementary, and both are integral to the efficacy of the proposed DGT.

4) Composite Feature Fusion Learning: In this experiment, the CFFL module is removed from the proposed DGT, which means that the final extracted features are complex, deep, and encapsulated global attributes. As observed from the experimental results in Fig. 14, relying solely on global and complex deep features leads to diminished performance. This decline is attributed to the absence of the CFFL, which typically integrates the abstract global information extracted by the deep layers of the DGT with detailed local information from various layers. This integration enhances the discriminative power of the final features. Moreover, the extent of performance degradation observed from removing the CFFL ranks just below that of removing the DGT, further underscoring the significance of this module.

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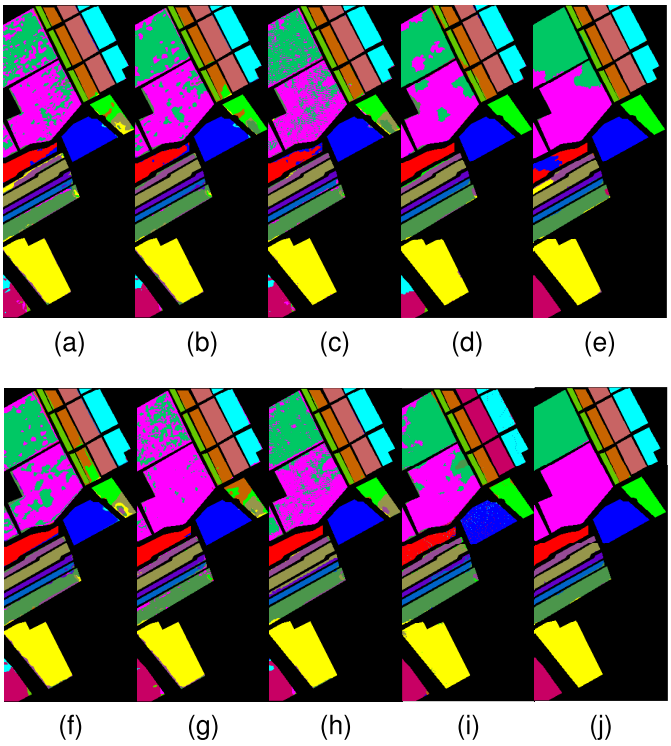


Fig. 11. Classification maps obtained by different methods on the Salinas dataset. (a) 3DCNN. (b) DFFN. (c) RSSAN. (d) GT-all. (e) GT-adj. (f) SF. (g) SSFTT. (h) GAHT. (i) GraphGST. (j) DGT.

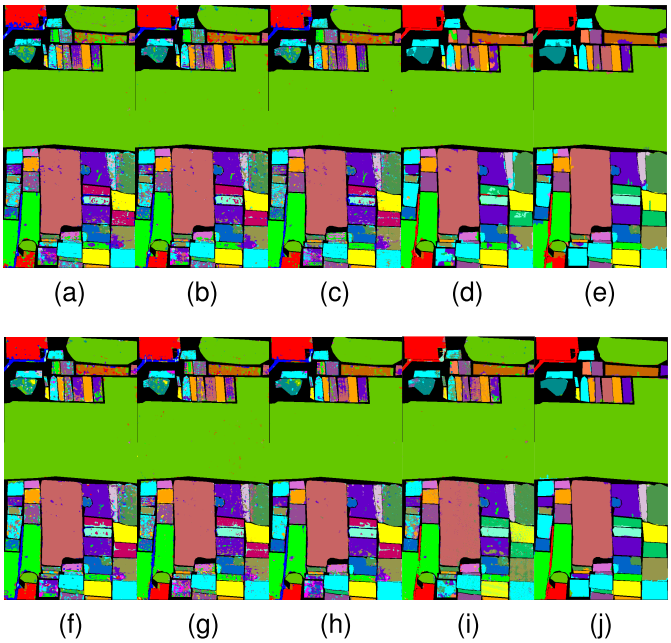


Fig. 12. Classification maps obtained by different methods on the WHU-Hi-HongHu dataset. (a) 3DCNN. (b) DFFN. (c) RSSAN. (d) GT-all. (e) GT-adj. (f) SF. (g) SSFTT. (h) GAHT. (i) GraphGST. (j) DGT.

B. Influence of Label Ratio

To assess the robustness of the proposed DGT, experiments are conducted across four datasets using varying training sample sizes, and the results are compared against other algorithms. Specifically, 1%–5% of the labeled samples are randomly selected as training samples for the Indian Pines dataset and 0.1%–0.5% for the Salinas, HongHu, and YRE

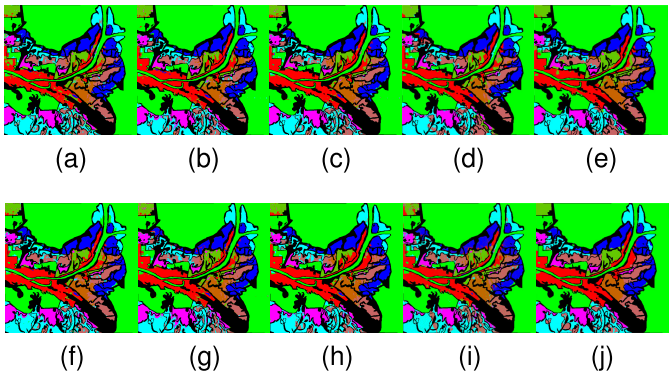


Fig. 13. Classification maps obtained by different methods on the YRE dataset. (a) 3DCNN. (b) DFFN. (c) RSSAN. (d) GT-all. (e) GT-adj. (f) SF. (g) SSFTT. (h) GAHT. (i) GraphGST. (j) DGT.

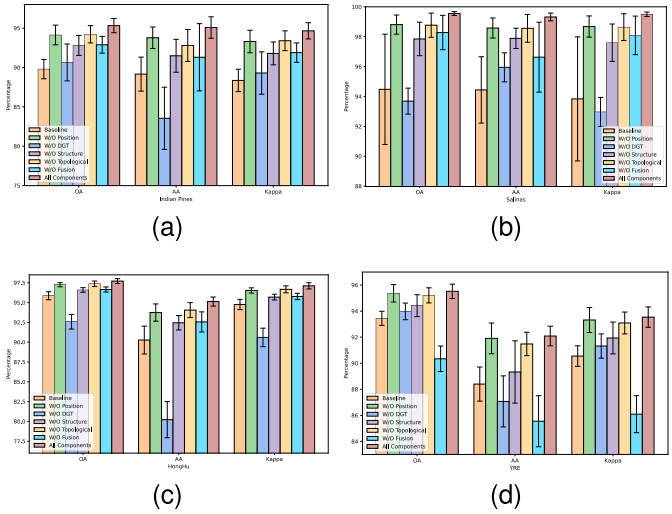


Fig. 14. Ablation analysis of different components in the proposed model on four datasets. (a) Indian Pines. (b) Salinas. (c) WHU-Hi-Honghu. (d) YRE.

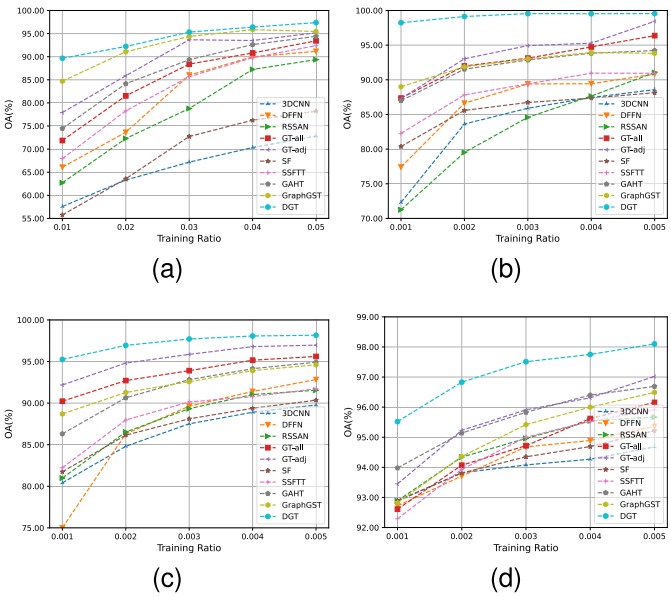


Fig. 15. Classification performance of each method with different training ratios on four datasets. (a) Indian Pines. (b) Salinas. (c) WHU-Hi-Honghu. (d) YRE.

datasets. Fig. 15 details the OA values for DGT alongside other comparative algorithms. It is evident that the best

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TABLE XII
RUNNING TIME (S) OF DIFFERENT METHODS ON FOUR DATASETS

Dataset	Time	3DCNN	DFFN	RSSAN	GT-all	GT-adj	SF	SSFTT	GAHT	GraphGST	DGT
Indian Pines	Train time(s)	48.38	120.94	74.51	11.23	10.39	180.53	61.11	100.57	86.45	43.2
	Test time(s)	1.54	5.2	3.31	0.75	0.72	27.8	2.4	3.47	24.38	1.19
	Total time(s)	49.92	126.14	77.82	11.98	11.11	208.33	63.51	104.04	110.83	44.39
Salinas	Train time(s)	28.2	53.84	39.71	12.51	11.55	198.22	33.15	49.45	25.72	78.77
	Test time(s)	27.2	31.85	51.38	2.7	3.11	106.31	20.54	67.02	53.46	4.01
	Total time(s)	55.4	85.69	91.09	15.21	14.66	304.53	53.69	116.47	79.18	82.78
WU-Hi-HongHu	Train time(s)	215.92	433.03	295.12	97.47	97.45	2436.81	264.84	415.75	214.14	195.83
	Test time(s)	26.27	87.09	45.39	67.94	69.01	953.43	34.56	73.3	264.88	84.18
	Total time(s)	242.19	520.12	340.51	165.41	166.46	3390.24	299.4	489.05	479.02	280.01
YRE	Train time(s)	97.76	136.24	97.95	77.24	76.21	440.06	107.10	189.44	73.28	129.69
	Test time(s)	34.71	89.12	59.09	36.93	38.76	1643.82	34.18	56.45	132.52	34.95
	Total time(s)	132.47	225.36	157.04	114.17	114.97	2083.88	141.28	245.89	205.80	164.64

classification performance for all methods is achieved at the highest training ratios. As the number of training samples decreases, each method exhibits varying degrees of performance decline. Notably, the DFFN, RSSAN, SF, and SSFTT algorithms show the most significant reductions in the Indian Pines dataset. In contrast, on all four datasets, the proposed DGT maintains high performance even with fewer training samples, demonstrating superior robustness under limited sample conditions.

C. Comparison of Running Time

The running times of the different methods on the four datasets, including training time, testing time, and total time, are presented in Table XII. As shown in Table XII, GT-all and GT-adj exhibit the shortest running times across the four datasets, which can be attributed to their simple network structures. This is because the oversmoothing issue becomes severe as the network depth increases. Other transformer-based methods require more time than CNN-based methods. For the proposed DGT, the training time is longer compared to CNN- and transformer-based methods. However, its running time is comparable to that of CNN-based methods and lower than that of other transformer-based methods. Compared to the performance improvement achieved, the increase in computational costs is tolerable.

D. Influence of Sample Locations

To explore the impact of training sample locations on the performance of the proposed DGT, experiments with fixed training samples are conducted on the four datasets. The sampling interval of the training samples ranges from 100 to 1000, and the results are shown in Fig. 16. It can be observed that the performance of DGT is poor when the sampling interval is small. As the sampling interval increases, the performance of DGT on large datasets increases dramatically and stabilizes when the sampling interval reaches 300 or more. This indicates that a broader distribution of training samples across the dataset allows DGT to learn more positional information, and the performance improvement from this positional information will eventually reach a saturation point.

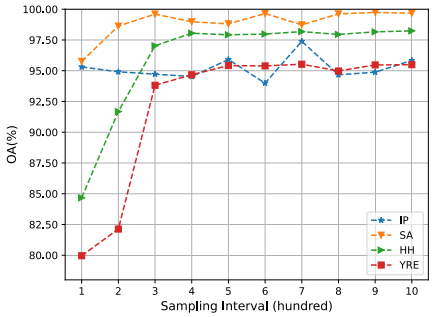


Fig. 16. Performance of DGT at different sampling intervals.

E. Limitations and Future Work

For the proposed method, the input is HSI encoded as a graph structure. This implies that as the dataset size increases, the graph structure becomes larger, resulting in higher computational costs for the model. Although most of the HSI classification datasets can be processed by the DGT, this places higher demands on the hardware used for model training. In addition, the deformed positional space only includes spatial information, which limits the performance of DGT to some extent. This also implies that after mapping the graph nodes into the positional space, the proposed DGT can be used to handle graph-structured data. In future research, the impact of small batch graphs will be explored to reduce the computational cost, and the spatial and spectral features will be simultaneously considered to generate a high-dimensional positional space for more discriminative location information.

V. CONCLUSION

In this article, a graph transformer framework named DGT is proposed for the HSI classification task. This framework comprises three main components: the LAPE module, the DGT layer, and the CFFL module. The method leverages the global modeling capabilities of the transformer along with the advantages of GNNs in processing irregular graph data, addressing the challenge of capturing long-range global information that is often problematic for methods based solely on transformers or GNNs. Specifically, the LAPE module initially encodes the HSI into a graph structure and generates positional

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embeddings for each graph node, enabling the transformer to recognize the positional information of different regions. Then, the DGT layer, which integrates graph structure encoding and graph topology encoding, is meticulously designed to capture both local structural relationships and long-range topological dependencies among nodes composed of ground objects by sampling virtual neighbors across various ranges and distances. After stacking multiple layers of DGT, the model extracts features ranging from local to global and from simple low level to complex high level, which are then input into the CFFL module to enhance the discriminative power and expressive capability of pixel features. Extensive experiments on four standard datasets demonstrate that the proposed DGT significantly outperforms other advanced algorithms in feature extraction capability and achieves SOTA classification performance.

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Yingjie Lu received the B.Eng. degree in electronics and information engineering from Northwestern Polytechnical University, Xi'an, China, in 2022, where he is currently pursuing the M.Eng. degree in information and communication engineering with the School of Electronics and Information.

His main research interests include remote sensing and deep learning.



Xiaofei Wang (Graduate Student Member, IEEE) received the B.Eng. degree in electronics and information engineering from Northwestern Polytechnical University, Xi'an, China, in 2022, where she is currently pursuing the M.Eng. degree in information and communication engineering with the School of Electronics and Information.

Her main research interests include remote sensing and image processing.



Shaohui Mei (Senior Member, IEEE) received the B.Eng. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

He is a Professor with the School of Electronics and Information, Northwestern Polytechnical University. He was a Visiting Student at The University of Sydney, Sydney, NSW, Australia, from October 2007 to 2008. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Dr. Mei received the Excellent Doctoral Dissertation Award of Shaanxi Province in 2014, the Best Paper Award of the IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS) in 2017, the Best Reviewer of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS) in 2019, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) in 2022. He served as an Associate Editor for IEEE TGRS and IEEE JSTARS, and a guest editor for several remote sensing journals. He also served as the Registration Chair of the IEEE China Summit and International Conference on Signal and Information Processing (ChinaSIP) in 2014.



Fulin Xu received the B.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2021, where he is currently pursuing the master's degree in signal and information processing.

His research interests include hyperspectral remote sensing image processing and deep learning.



Mingyang Ma (Graduate Student Member, IEEE) received the B.Eng. degree in communication engineering and the Ph.D. degree in communication and information systems from Northwestern Polytechnical University, Xi'an, China, in 2015 and 2021, respectively.

His main research interests include video summarization and image processing.

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